EuroSafeGuard: Open Constitutional Classifiers

Sid's reading notes

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Black text follows my handwritten notes. Blue text marks LLM recommended additions.

Abstract

- **Motivation:** state-of-the-art open-weight safeguards are inference-expensive; state-of-the-art multi-stage constitutional classifiers are closed source. The authors introduce multi-stage constitutional classifiers to solve these problems.
- **Contributions:** a constitution covering 11 harm categories; data generated from it; a faithful recreation of CC++ (EuroSafe-CC++); and a cascade using lightweight probes, an input classifier and an exchange classifier.
- They compare with 16 open guard models using a uniform false-positive rate on real traffic where possible; gpt-oss-safeguard uses its native verdict.
- When in front of Llama-3-8B, EuroSafeGuard reduces attack success on held-out Jailbreak Foundry attacks from 19.7% to 5.4% (under StrongREJECT), and blocks 0.2% of WildChat-clean prompts.
- The probe + input classifier catches the most attacks on three of four held-out jailbreak sets. PAIR breaks about 21% of behaviors through EuroSafeGuard's input stage, versus 49% through HaloGuard, without the guard in the search loop.
- The cascade costs approximately 15% extra in compute, averaged across queries. However, this assumes the probe as free. This works under the assumption the probe is reading activations already computed by the model being protected.

1 Introduction

- Constitutional classifiers, CCs for short, are guardrails: they evaluate harmful intent and whether a reply gives harmful assistance.
- CCs are trained on synthetic data generated from a constitution, i.e. a list of harmful and benign behaviors/content categories.
- Unlike Claude's constitution or OpenAI's Model Spec, which describe how a model should behave, the constitution here defines what the guard blocks and allows.
- CC++, Anthropic's latest closed source constitutional classifier, was tested on top of an in-house Claude model that had already undergone AI safety training. Therefore, verifying the guard's contribution independently is not possible in the same way as with an open model.
- This paper introduces EuroSafe-CC++, which follows the CC++ design as closely as possible. It then improves on this design, introducing EuroSafeGuard. The guards are calibrated on real chat traffic.

Introduction, pp. 1–3; taxonomy, p. 6.

3 Problem formulation and threat model

Guardrails

- An LLM, M, has a constitution, C. We write $G(M)$ for the guarded system: that is, a guardrail, G, wrapping M.
- For an input x, a part of G decides whether M may answer; simultaneously while M prepares a reply y, another part of G may stop this part of the process too.
- G consists of a probe, P, an input classifier, C_I , and an exchange classifier, C_E, alongside cascade routing and decision rules.
- The probe, P, is a small classifier on hidden activations (from Llama-3.1-8B-Instruct in these experiments, for every protected model). P scans x and determines if it should be elevated to the classifiers or M should simply respond to the P judged as unharmful query.
- In the intended deployment, the probe P is cheap, since the activations already exist given M. Most traffic only incurs the cost of P, so the system remains cheap on most traffic.
- If escalated by P, C_I then reads x before the reply, contributing to the final combined score.
- If x is escalated by P, C_E reads x and the reply y. Its score is combined with those of P and C_I to decide whether y is shown or blocked.

Threat model

- The attacker wants a reply, y, that meaningfully helps with a behavior the constitution forbids. The guardrail weights are public.
- Attackers can query the guarded system $G(M)$ many times. Note, queries and optimisation steps presented in this paper are bounded.
- The jailbreak attacks they test G against are:
- **Held-out jailbreaks:** apply attack templates and transformation techniques to behaviors not seen in the training dataset (some templates are seen).
- **Search attacks:** iterate x until the model complies with the harmful request, without seeing the guard.
- **Adaptive attacks:** search using the guard's decision on each x and use this to update attack prompts.
- **White-box:** the attacker has gradient access to M, P and C_I. The attacker cannot change the deployed weights.

Operating point

A guard can always block more attacks by flagging more requests: a so-called recall over-refusal trade-off. For this reason, they generally compare guards at the same false-positive budget: this being the flagging of 5% of benign WildChat-clean prompts. gpt-oss-safeguard uses its native verdict; the deployed cascade instead uses caps for individual benign benchmarks. WildChat-clean is real chat traffic with harmful requests removed.

Metrics

- **Recall** $\uparrow$: share of harmful prompts the guard flags at the 5% budget.
- **Behaviors broken** $\downarrow$: percentage of harmful goals for which there is at least one harmful actual reply delivered.
- **Attack success** $\downarrow$: share of individual attack prompts that generate a harmful y and have it delivered.

- **Benign failure** ↓: share of poorly handled benign requests: blocked, refused or answered with broken text.
- **Cost** ↓: computation added per request, as a share of the model's FLOPs (floating-point operations).

Section 3, pp. 5–6.

4 Method

4.1 Taxonomy

The constitution covers 11 harm categories. Each behaviour included across these 11 harm categories is labelled as harmful, dual-use harmful, dual-use benign or benign, with a further group of general benign requests.

4.2 Data

Constitution

They use the BELLS-O constitution, generated by CeSIA using Claude Opus 4.6. It has about 5k behaviours (4,904 entries). Dual-use labels are given to data that can be both benign or harmful depending on context; these are grouped into the respective classes. Removing 46 harmful entries on sexual content involving minors leaves 4,858 entries: roughly 45.5% harmful and 54.5% benign.

Type	Entries
Harmful	1,172
Dual-use harmful	1,039
Dual-use benign	1,084
Benign	1,443
General benign	120
Total	4,858

Prompt generation

To generate 98k of the 540k prompts used to train the classifier, the following approach is used:

- REDACT expands each entry using an uncensored generator; Huihui-Qwen3.8-27B-abliterated. This creates both short prompts of 5–20 words and long prompts of 2–5 sentences.
- A checker, also the Huihui-Qwen3.8-27B-abliterated but given different instructions, that knows the target level accepts, regenerates or rejects each prompt.
- Then these 5k behaviors become 17,903 accepted request prompts. This is through requesting three short and three long prompts per behaviour (also using the Huihui-Qwen3.8-27B-abliterated, again with a specified system prompt).
- REDACT also removes model fingerprints: this is an automated process of paraphrasing generated prompts to make the AI-generated text less recognisably model-written. The paper specifically describes this for BELLS-O evaluation data.
- Then, for each entry in the $17,903$ dataset, paraphrasing, natural variations and jailbreak rewrites take the dataset to about 98k accepted rows, including originals. The exact accepted split is not available locally. The four-round schedule comes from REDACT's configuration, combined with EuroSafeGuard's recipes.

They report on the remaining data used to train the classifiers in Appendix D; the table below groups these sources:

Training data

Source	Rows
REDACT originals, paraphrases, natural variations and jailbreak rewrites	about 98k
Jailbreak Foundry attacks on them	about 11k
CeSIA constitution-input prompts	26,979
Further obfuscation, DAN and Best-of-N augmentations	about 53k
Other safety and benign data (rounded remainder)	about 352k
Total	**540,536**

Moreover, they explicitly state three principles used to construct the training data:

1. **Every augmentation is applied to benign text too.** Otherwise the guard could learn to flag the wrapper rather than the underlying request.
2. **Every attacked prompt has a plain twin.** For every attack wrapped in a role-play obfuscation, the attack also has its plain, non-role-play version also within the dataset. A consistency term pulls their representations together. That is, they compare the classifier's last-token hidden vectors using cosine similarity. Therefore, the rewritten prompt's representation is encouraged to align with the original's.
3. **Each generator contributes both classes.** For exchange-classifier training, one needs both the input prompt data but also generated responses, y from the model M. Their reply generators include three variants on M: an undefended Llama-3-8B-Instruct, a circuit breaker variant, and a triplet training variant [1]. Each contributes both harmful and harmless replies. The reason being, these distinctions all produce responses, y, that contain signatures of which M variant they are. If the variants differ in how harmful the (x, y) pairs they generate are, the classifier may accidentally learn this correlation. In fact, they initially found this to be the case: their classifiers learned to distinguish between models based on the characteristics of their responses, rather than solely assessing the harmfulness of those responses as intended. If left unaddressed, this could limit the guard's usefulness in production settings, where clients may deploy the guard alongside any open-weight model of their choosing.

The judge used to assess generated replies, y, was gpt-oss-safeguard-20b, supplied with the behaviour-specific GuidedBench rubric. A reply was deemed harmful if the judge assigned it a score of ≥ 0.5. Labels from this judge were used to train part of the exchange-classifier dataset.

Section 4.2, p. 7; Appendix D, p. 35.

4.3 EuroSafe-CC++: A Baseline Recreation

A faithful open recreation of CC++. It uses lightweight prompt and reply probes, then repurposes an input-only classifier to read the exchange. This is the baseline for the modified EuroSafeGuard design.

[1] Undefended: the original Llama-3-8B-Instruct, retaining its existing safety training but with no added experimental defense.
Circuit-breaker defended: the model is further trained to redirect internal representations associated with harmful replies, disrupting their generation.
Triplet defended: the model is further trained using a triplet loss that separates harmful and benign internal representations to improve resistance to jailbreaks

4.4 EuroSafeGuard: A Multi-Stage Constitutional Guardrail

$$x \longrightarrow M \longrightarrow y$$

Llama-3.1-8B-Instruct's activations $\longrightarrow P \longrightarrow$ escalate suspicious requests

Escalated: $C_I(x)$ and $C_E(x, y) \longrightarrow$ combined score $\longrightarrow$ show / block

- Using the input classifier to judge exchanges performs worse than using a classifier trained on exchanges.
- **Alert mode:** instead of blocking every flagged non-CBRN request, the system note names the triggered policy. The model can answer; C_E checks the reply more strictly. Requests flagged as prohibited CBRN remain blocked.

The deployed cascade escalates the top **10%** on benign calibration traffic (Section 4.5, p. 8). This is separate from the 5% false-positive comparison budget. Its score is

$$s_G = 0.6\, z_P + 0.2\, z_{C_I} + 0.2\, z_{C_E},$$

where each z is the standardized score. This weighted decision score is separate from the definition of G as the complete guardrail.

5 Experimental setup

Benchmarks

- BELLS-O: **1,100 harmful and 300 benign prompts**, made by safety researchers, existing benchmarks and adversarial generation. Separate from training data.
- PyRIT jailbreak templates; BELLS-O in-house jailbreaks; BET; GuidedBench $\times$ Jailbreak Foundry. Count the successful attacks the unguarded model answers harmfully.

Judges

gpt-oss-safeguard-20b applies a GuidedBench behavior-specific rubric. A score ≥ 0.5 counts as harmful. StrongREJECT provides a second check. "Strictly harmful" requires StrongREJECT ≥ 0.8 and the safeguard judge's harmful verdict.

Baselines

Compare with 16 open guard models.

Protocol

- Guards are generally scored in input mode at the 5% WildChat-clean budget.
- PAIR/TAP: run against an unguarded model, score attempts per guard; a behavior is broken if any harmful attempt passes.

Attacks with the guard in the loop

The attacker now queries the guarded system. Blocked attempts count as failures. PAIR sees blocking feedback but is not designed specifically to evade a guard. Diffusion and white-box attacks target the guard itself more directly.

Configurations of EuroSafeGuard

- **Input:** probe and input classifier.
- **Input+Exchange:** also checks the reply. The stricter in-loop experiment uses a 5% input budget and a 1% exchange threshold.
- **Deployed:** the cascade with its combined score; blocks about 0.2% of WildChat-clean.

Training runs

Results are generally averaged over five seeds.

Identical training data with different seeds leaves **11–48% of PAIR behaviors broken**; a single run can mislead (Section 6.4; Table 23).

Section 5, pp. 9–11.

6 Results

Section 6.1 (RQ1). Tests whether the input stages detect attacks they were not trained on.

- EuroSafeGuard's input stage has the highest recall on three of four held-out jailbreak sets.

Section 6.2 (RQ2). Tests whether the guardrail holds when attackers use feedback or gradients to adapt to it.

- The activation probe supplies much of the robustness against these attacks.

Section 6.3 (RQ3). Measures the cost to benign users and the extra computation required.

- Alert mode with the strict exchange threshold reduces refusal/blocking of red-team benign requests from 37% to 5–16% across six protected models.

7 Limitations

Attacks with the guard in the loop

For 100 PAIR behaviors, strictly harmful replies get through for about:

- No guard: **26/100**.
- EuroSafeGuard, stricter Input+Exchange setting: **5/100**.
- EuroSafeGuard, deployed: **20/100**.
- HaloGuard, at the 5% budget: about **21/100**.

An attacker with gradients bypasses the deployed cascade's **input scores on 40% of behaviors**; that does not mean 40% harmful replies delivered. Reconstruction and output obfuscation remain concerns. Adaptive training and human red teaming are next steps.

Protected models

Most end-to-end tests use Llama-3-8B-Instruct. gpt-oss-20b is tested end to end and in alert mode, but not with guarded-system feedback. The probe only reads Llama-3.1-8B-Instruct, including when another model supplies the reply. It is not run on gpt-oss activations; a probe on that model's activations would need training.

Policy breadth

Specialist models still outperform this approach on general content moderation.

Sensitive benign requests

Difficult benign requests are flagged: refusing all flags rejects 37% of red-team benign requests. This falls to 5–16% in alert mode with the strict exchange threshold, across six protected models. It is a reduction in rejections, not necessarily in flags. At most 3.6% of chemical/biological goals are broken in that alert-mode experiment.

Calibration

Thresholds use the first benchmark halves, but weights and escalation settings were chosen using constraints on both halves. Benign rates near the caps may therefore be optimistic. Attack-test behavior halves were untouched.

Judges

LLM judges can misjudge adversarial edge cases. A second judge helps check results but does not eliminate that uncertainty.

Section 7, pp. 17–18; Tables 4, 13 and 16.

Ethics statement

- Open weights help both researchers and attackers. Robustness researchers need the weights.
- Training prompts require gated access. Training replies are not public: shared for safety research on request, after manual review.
- Protection has costs and limits. It lowers misuse risk but does not remove it; false positives affect benign users.

Ethics statement, p. 18.

Further reading: optional omissions

Added reading guide; not in handwritten notes.

- Training details: Table 5 and Appendix E; training/README.md for soft labels, twin loss and sampling.
- Calibration: Section 4.6 and Appendix F.3.
- Multi-turn and low-resource-language tests: Section 6.1 and Appendix F.1.
- Full attack results and seed spread: Appendix F, Tables 12–13, 20 and 23.
- Reproduction limits: data/README.md; no supplied script rebuilds the final 541k mixture row for row.